

Data Mining Numerical Problems & Solutions

CT 72502 (Elective I) · BEI / BEX IV-I · TU, IOE · Chapter 6: Anomaly / Fraud Detection

Ch 6 – Anomaly / Fraud Detection – Numerical Problems & Solutions

3 numerical marks, 0.2% of the archive, 2% of this chapter · This chapter is **almost entirely theory**. Across all 23 papers there is exactly **one** numerical 76 Ch, on the base rate fallacy. It is solved below, plus 2 practice problems covering the patterns a future numerical would most likely take.

Methods

- **M1** Base rate fallacy: detection rate, false alarm rate, Bayesian detection rate
- **M2** Distance-based outlier: the $DB(r, \pi)$ test

76 Ch
practice

Every problem below ends with its solution drawn step by step. Each panel carries a chip naming the move it makes; the decisive step is framed **KEY STEP**. The table says what each move is and where it is used.

The moves every chapter-6 numerical below is built from

Move	What it says	Where it is used (problem, step; orange = the key step)
round population	Invent a round population and count objects; no formula to misremember.	1.1 1; Practice A 1
apply both rates	Detection rate hits the anomalies, false alarm rate hits the much larger normal group.	1.1 2; Practice A 2
read the alarm bar	$P(I A)$ = true alarms / all alarms, which is what an analyst sees.	1.1 3 ; Practice A 3
base rate moved	Same detector, rarer anomalies: only the prior changed, and $P(I A)$ collapses.	1.1 3 ; Practice A 3
count within r	Count the objects within r , the point itself included.	Practice B 1
compare with π	Outlier when that count over $ D $ is at most π .	Practice B 2
k -NN cross-check	With $k = \lceil \pi D \rceil$, outlier iff the k -th neighbour is farther than r .	Practice B 3

M1. Base rate fallacy

Method

- Let I = anomalous, $\neg I$ = normal, A = alarm raised.
- **Detection rate** = $P(A|I)$ **False alarm rate** = $P(A|\neg I)$
- **Bayesian detection rate** (what an analyst actually cares about):

$$P(I|A) = \frac{P(A|I) P(I)}{P(A|I)P(I) + P(A|\neg I)P(\neg I)}$$

- **Fastest safe route**: invent a round population (10,000), build the confusion matrix, read every rate off it. No formula to misremember.

Problem 1.1 3

QUESTION AS PRINTED 2076 Chaitra · Q11 (b) · [3]

If the probability that a normal object is classified as an anomaly is 0.01 and the probability that an anomalous object is classified as anomalous is 0.99, then what is the false alarm rate and detection rate if 99% of the objects are normal?

Given

- $P(A|\neg I) = 0.01$ (normal object raises an alarm)
- $P(A|I) = 0.99$ (anomalous object raises an alarm)
- $P(\neg I) = 0.99$, so $P(I) = 0.01$

Step 1: the two rates asked for, read straight off the givens

- False alarm rate = $P(A|\neg I) = 0.01$ (1%)
- Detection rate = $P(A|I) = 0.99$ (99%)

Step 2: why the 99% prior was given – build the matrix on 10,000 objects

	Alarm	No alarm	Total
Anomalous (I)	$0.99 \times 100 = 99$ (TP)	1 (FN)	100
Normal ($\neg I$)	$0.01 \times 9900 = 99$ (FP)	9801 (TN)	9,900
Total	198	9,802	10,000

- Check: detection rate = $99/100 = 0.99$ ✓ false alarm rate = $99/9900 = 0.01$ ✓

Step 3: Bayesian detection rate

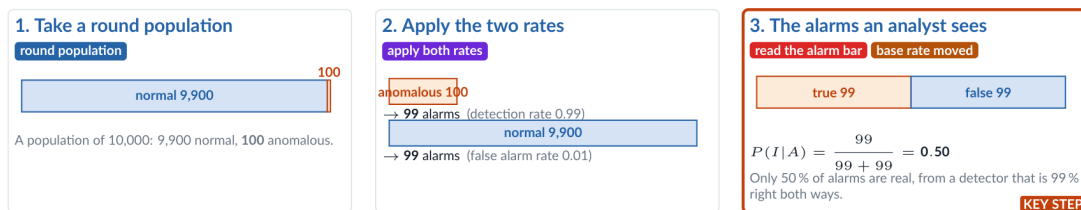
$$P(I|A) = \frac{(0.99)(0.01)}{(0.99)(0.01) + (0.01)(0.99)} = \frac{0.0099}{0.0099 + 0.0099} = \frac{99}{198} = 0.50$$

Final answer

- False alarm rate = **0.01**, detection rate = **0.99**.
- **But the Bayesian detection rate is only 0.50** – half of every alarm raised is false, despite a detector that is 99% accurate in both directions.
- **This is the base rate fallacy.** The normal population (9,900) is so much larger than the anomalous one (100) that its 1% error rate manufactures exactly as many false alarms (99) as there are true detections (99).
- **Say this in the exam:** it is the whole point of the question and is worth more than the two arithmetic lines.

If the examiner wanted only the two conditional rates, they are given directly and the question is trivial – which is the clue that the **99% prior** must be used for something. Answer both readings: state the two rates, then compute $P(I|A)$ and name the fallacy. verified/added

Step by step: why a 99% detector leaves half its alarms false



Practice A – rarer anomalies

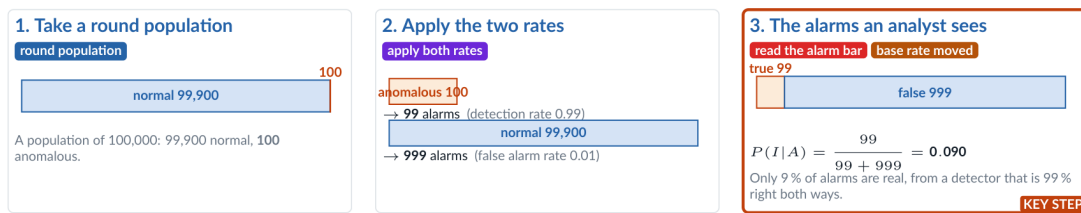
PROBLEM not a PYQ · same detector, rarer anomalies verified/added

The detector of Problem 1.1 is unchanged: a normal object is classified as an anomaly with probability 0.01, and an anomalous object is classified as anomalous with probability 0.99. Now only **1 object in 1,000** is anomalous instead of 1 in 100. Find the detection rate, the false alarm rate, and the probability that an object raising an alarm really is an anomaly.

Shows how fast the Bayesian rate collapses. verified/added

- On 100,000 objects: 100 anomalous, 99,900 normal.
- $TP = 0.99 \times 100 = 99$ $FP = 0.01 \times 99,900 = 999$
- $P(I|A) = \frac{99}{99 + 999} = \frac{99}{1098} = 0.090$
- **Only 9% of alarms are real.** Detection rate and false alarm rate are unchanged at 0.99 and 0.01 – **only the base rate moved.**
- **Conclusion (Axelsson):** to keep $P(I|A)$ usable when $P(I)$ is tiny you must drive the *false alarm rate* down, not push the detection rate up.

Step by step: the same detector, anomalies ten times rarer



M2. Distance-based outlier: the $DB(r, \pi)$ test

Method

- Object o is a $DB(r, \pi)$ -outlier if

$$\frac{|\{o' \mid \text{dist}(o, o') \leq r\}|}{|D|} \leq \pi$$

- In words: count how many objects (including o itself, per Han & Kamber) lie within distance r ; if that count as a fraction of $|D|$ is at most π , it is an outlier.
- Equivalent form: with $k = \lceil \pi |D| \rceil$, o is an outlier if $\text{dist}(o, o_k) > r$.

Practice B

PROBLEM not a PYQ · written to fill the gap **verified/added**

A dataset contains the points $P_1(1, 1)$, $P_2(1, 2)$, $P_3(2, 1)$, $P_4(2, 2)$ and $P_5(8, 8)$. Taking $r = 2$ and $\pi = 0.4$, and using Euclidean distance, determine which points are $DB(2, 0.4)$ -outliers.

No PYQ has set this yet, but it is the only computable form of the distance-based question that is asked in 6 papers, so it is worth one run-through.

verified/added

Step 1: distance matrix

	P_1	P_2	P_3	P_4	P_5
P_1	0	1	1	1.414	9.899
P_2	1	0	1.414	1	9.220
P_3	1	1.414	0	1	9.220
P_4	1.414	1	1	0	8.485
P_5	9.899	9.220	9.220	8.485	0

Step 2: count neighbours within $r = 2$

(threshold $\pi|D| = 0.4 \times 5 = 2$ objects)

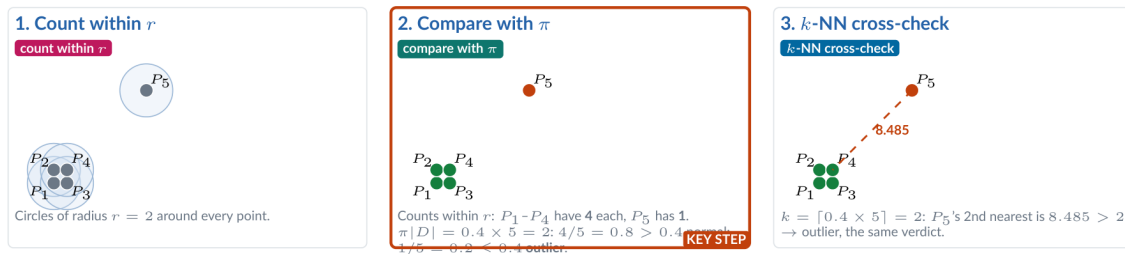
- P_1 : itself, $P_2, P_3, P_4 = 4$ $4/5 = 0.8 > 0.4 \rightarrow$ normal
- P_2, P_3, P_4 : 4 each by symmetry $\rightarrow$ normal
- P_5 : only itself = 1 $1/5 = 0.2 \leq 0.4 \rightarrow$ **outlier**

Final answer

P_5 is the only $DB(2, 0.4)$ -outlier.

- Cross-check with the k -NN form:** $k = \lceil 0.4 \times 5 \rceil = 2$. For P_5 , its 2nd nearest neighbour is at $8.485 > 2 \rightarrow$ outlier $\checkmark$. For P_1 , its 2nd nearest is at $1 \leq 2 \rightarrow$ normal $\checkmark$.
- Note the sensitivity to r :** raise r to 9 and P_5 stops being an outlier altogether. Quoting the chosen r and π is part of the answer.

Step by step: the DB(2, 0.4) test, and the k -NN form of the same test



Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
DM	data mining	DW	data warehouse
DB	database	DBMS	database management system
KDD	knowledge discovery in databases	DSS	decision support system
OLTP	on-line transaction processing	OLAP	on-line analytical processing
ROLAP	relational OLAP	MOLAP	multidimensional OLAP
HOLAP	hybrid OLAP	ETL	extract, transform, load
BI	business intelligence	ER	entity-relationship
attr(s)	attribute(s)	dim(s)	dimension(s)
multidim	multidimensional	txn(s)	transaction(s)
info	information	ops	operations
repo	repository	freq	frequent / frequency
config	configuration	no.	number
PCA	principal component analysis	SVD	singular value decomposition
SMC	simple matching coefficient	KNN	k -nearest neighbour
ANN	artificial neural network	POS	point of sale
FK	foreign key	UI	user interface
GIGO	garbage in, garbage out	w/, w/o	with, without
rm	remove	vs	versus
Eg	example / for example	+Fig	draw a figure
→	leads to / therefore	+	and

Exam-paper month codes (used in the year tags): Ba = Baishakh · Jth = Jestha · Asa = Ashar · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Mng = Mangsir · Po = Poush · Ma = Magh · Ch = Chaitra.

Tier chips: **TOP /23** = topic asked in 7 or more of the 17 papers · **HOT /23** = 4–6 papers · **PIN /23** = 1–3 papers but worth ≥ 8 marks. **Mark chips** such as **2+5** are the marks that question actually carried. **verified/added** marks anything researched and added that was not in the source notes.